Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_075148_r
- started_at_local: 2026-05-20 07:51:48
- finished_at_local: 2026-05-20 08:22:42
- duration_hhmm: 00:31
- duration_seconds: 1853.542
- seed_offset: 17000
- replicate_label: r
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.100187	0.0	0.063755	0.692044	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.843989	0.4375	False	0.008128	0.938393
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.046321	0.115856	0.896926	0.4425	False	0.00879	0.909128
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235667	0.054968	0.121542	0.783221	0.44	False	0.004996	0.384599

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235667	0.054968	0.121542	0.0	0.4375	False	0.006824	0.398175
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.0	0.4425	False	0.005954	0.227707

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.100187, family holdout 0.0, combined 0.063755.
- Accepted novelty 0.692044, complexity 0.76, and adaptation efficiency 0.027395.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.843989, complexity 0.4375, and adaptation efficiency 0.006857.
- Last-epoch runtime 99.643275 ms and distance evaluations 5266.75.
- Validation: surviving False, transplant delta 0.008128, positive scaffolds 1, Pareto runtime inflation 0.938393.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.881272, "gap_delta": -0.002615, "runtime_inflation": 0.938393, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.896926, complexity 0.4425, and adaptation efficiency 0.006452.
- Last-epoch runtime 60.1969 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.00879, positive scaffolds 1, Pareto runtime inflation 0.909128.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.88592, "gap_delta": -0.005462, "runtime_inflation": 0.909128, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.783221, complexity 0.44, and adaptation efficiency 0.007454.
- Last-epoch runtime 39.656 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.004996, positive scaffolds 1, Pareto runtime inflation 0.384599.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.289173, "gap_delta": 0.000224, "runtime_inflation": 0.384599, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.0, complexity 0.4375, and adaptation efficiency 0.0.
- Last-epoch runtime 126.971425 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.006824, positive scaffolds 1, Pareto runtime inflation 0.398175.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.308765, "gap_delta": 0.000224, "runtime_inflation": 0.398175, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 104.245025 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.005954, positive scaffolds 1, Pareto runtime inflation 0.227707.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.186363, "gap_delta": 0.000224, "runtime_inflation": 0.227707, "same_gap_faster": false}

Judge Appendix

markdown

TSP Modular-Operator Discovery Suite Review

Overview

- **Main transfer evidence:** held-out TSPLIB gap & family holdout gap.
- **Key criteria for interesting modular operators:** validation surviving transplant and ablation checks.
- **Prefer reusable operator structure claims over full-solver performance unless modular validation is strong.

- **No algorithm discovery claim:** no operator survives scaffold and family tests.

Summary of Conditions

Condition	Operator Type	Family Holdout Gap	TSPLIB Gap	Transplant Mean Gap Delta	Ablation Signal	Surviving Candidate	Comments Summary
phase7_baseline_heuristic_only	n/a	0.054968	0.235058	0.0	n/a	No	Baseline only; no novel operator.
phase7_full_solver_evolution	n/a	0.0	0.100187	0.0	n/a	No	Full-solver mode, no modular operator candidate surviving.
phase7_modular_operator_evolution	restart_controller	0.046321	0.235058	+0.008128 (worse)	Very weak (gap delta ~0.0026 w.r.t disabled)	No	Operator shows modest family gains on clustered instances but **fails transplant**, ablation weak; runtime inflated (~0.94).
phase7_modular_operator_random_replay	restart_controller	0.046321	0.235058	+0.00879 (worse)	Weak (gap delta ~0.0055 disabled)	No	Similar restart controller variant, same transplant/ablation failures and high runtime inflation (~0.91).
phase7_modular_operator_diversity_residual_replay	candidate_pruner	0.054968	0.235667	+0.004996 (worse)	Negative/Ablation shows no harm when disabled	No	Candidate pruner transfers poorly, no family improvements, ablation signal absent, runtime inflation moderate (~0.38).
phase7_modular_operator_compression_pressure	candidate_pruner	0.054968	0.235667	+0.006824 (worse)	Negative/Ablation shows no harm when disabled	No	Same limitations as above with compression pressure enforcement.
phase7_modular_operator_pareto_selection	candidate_pruner	0.054968	0.235667	+0.005954 (worse)	Negative/Ablation shows no harm when disabled	No	No family transfer benefit; runtime inflated (~0.23); no clean transplant.

Detailed Interpretation

Transfer & Held-out Gaps

- The baseline TSPLIB gap is high (~0.235).

- Modular operators (restart controllers) reduce family gap slightly on clustered families (approx. 0.0463 vs baseline 0.05497).
- Transfer gap improvements on held-out TSPLIB families are zero; no improvement in final TSPLIB gap across any modular operator.
- Candidate pruners do not improve family holdout or TSPLIB gaps.

Transplant Results

- Restart controllers **do not transplant cleanly** into other scaffolds (sparse_three_opt, clustered_local_search) and generally worsen mean gaps slightly (+0.008) on transplant.
- Candidate pruners never show meaningful positive transplant effects; gap deltas are slightly positive (worse) or near zero with no clear improvement.

Ablation Results

- Restart controllers have very weak ablation signals; disabling them results in only minor gap differences (~0.0026 to 0.0055), indicating limited operator impact.
- Candidate pruners show **no evidence of operator effect** since disabling does not harm performance, meaning their adaptive pruning likely does not contribute meaningfully.

Pareto & Runtime Tradeoffs

- Restart controller operators have **high runtime inflation** (~0.9) and distance evaluation inflation (~0.88) with only tiny gap improvements (less than 0.003).
- Candidate pruners have moderate runtime inflation (~0.23 to 0.38) with no gap improvement.
- No operator achieves better gap at same or faster runtime.

Survival & Selection

- No modular operator survives all tests (scaffold, transplant, family gap improvement, and ablation).
- All recovery signals negative or inconclusive.
- No positive scaffold count beyond 1; survival candidate flag is false for all.
- No operator qualifies as a robust reusable component based on provided data.

Conclusions

1. **No modular operator passes conservative acceptance criteria**:
- None improves held-out TSPLIB gap.
- None survives transplant tests with positive or neutral gap impact.
- Ablation tests suggest weak or no effective operator contribution.
- Runtime costs negate minor gap gains.

2. **Restart controllers show some family-specific benefit but fail transplant and ablation checks and have high runtime inflation**; they are **not reusable modular operators** in current form.

3. **Candidate pruner operators fail to improve family or TSPLIB gaps, show no ablation importance, and do not transplant cleanly**, making their core ideas unvalidated as modular operators.

4. **No claims for discovered algorithm or reusable operator structure are supported by this data**.

Recommendations

- Focus future exploration on improving operator transplant fidelity and ablation validation before claiming reusable operator structure.
- Investigate runtime inflation reduction for restart controllers to enable practical gains.
- Candidate pruner ideas may need reengineering or alternative adaptive criteria to demonstrate meaningful operator effect and transfer.

Final Statement

No modular operator candidate in this suite survives rigorous held-out TSPLIB gap, family holdout, transplant, and ablation validation criteria needed for reusable operator claims or algorithm discovery. The discovered restart controllers and candidate pruners require further development and validation before being considered reusable modular components for TSP solvers.